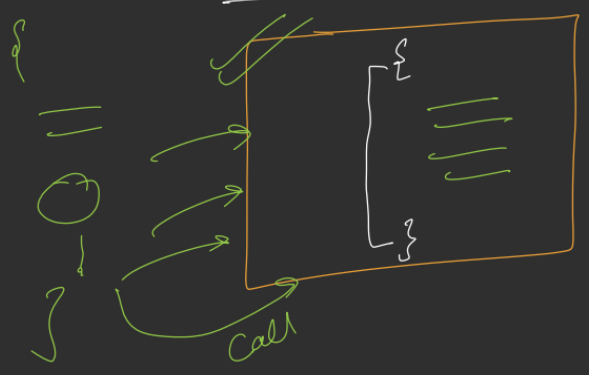
Functions in Java :

```
{
    int i = 4;
    int j = 5;
    int sum = i + j;
    System.out.println(sum);

    int k = 9;
    int l = 10;
    ____ (k+l);
}
```

Repeated code

Functions




```
Public static void main(String[] args){
    int i = 4;
    int j = 5;
    int result = sum(i, j);
    System.out.println(result);//9
}
```

Arguments

sum(10, 9); Calling a Function

```
Static int Sum(int a, int b){
    int result = a+b;
    return result;
}
```

Return Type

Parameters

Fn Name

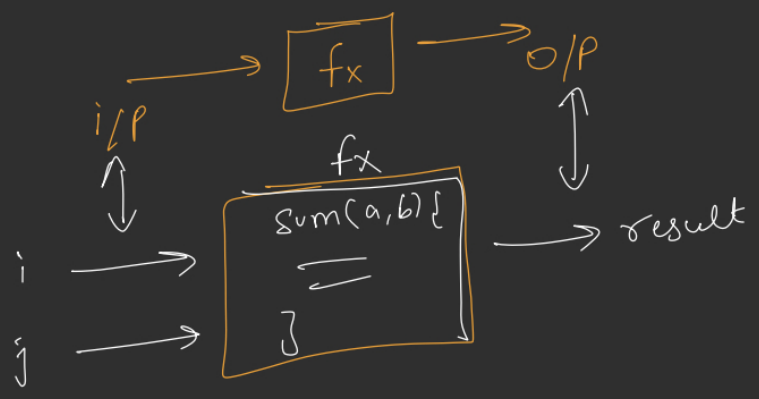
returning an output.

Return Type

⇓

Data type that is returned by fn.

Functions :



Types of Functions

- No I/P, No O/P
- I/P, No O/P
- No I/P, O/P
- I/P, O/P

Chaining of functions :

```
void main(——) {
    A();
}
```

Sys—— ("Bye");

```
void A() {
    B();
}
```

Sys—— ("Hi");

```
void B() {
    System.out.println ("Hello");
}
```

Chaining

Recursion

↳ A function which call itself.

```
main() {
    A();
}
```

```
void A() {
    A();
}
```

Base Case

```
main() {
    int n = 5;
    printNum(n);
}
```

```
void printNum(int n) {
    if(n==0) return;
    printNum(n-1);
    System.out.println(n);
}
```

10

main() {0}

printNum(5)
↳ printNum(4)
↳ printNum(3)
↳ printNum(2)
↳ printNum(1)
↳ printNum(0).

O/P

1
2
3
4
5

```
void printNum(int n) {
    if(n == 0) {
        return;
    }
    printNum(n-1);
    System.out.println(n);
}
```

Base Case

Recursive Call

$n \rightarrow 0^{th}$ 1st 2th 3th 4th 5th

fib $1, 1, 2, 3, 5, 8, \ldots$

Fibnoacci Series

n^{th} Fibonacci No.

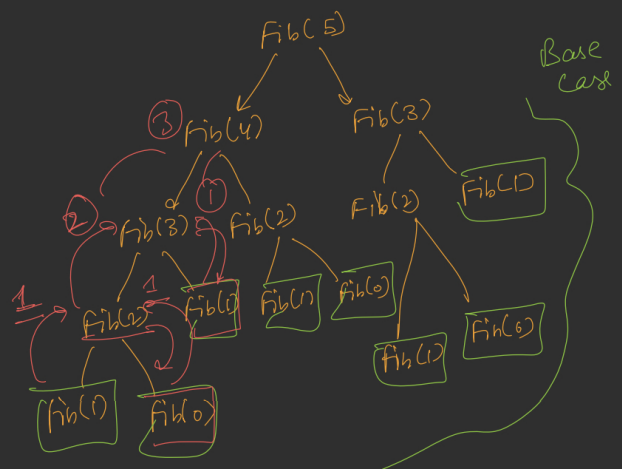
$\rightarrow 4$

$\rightarrow 5$

$$\boxed{Fib(n)} = \underbrace{\boxed{Fib(n-1)}}_{x} + \underbrace{\boxed{Fib(n-2);}}_{y}$$

```
int fib(n) {
    int x = fib(n-1);
    int y = fib(n-2);
    return x + y;
}
```

```
if (n==0 || n==1) {
    return 1;
}
```

1 —— 10

10 —— 1